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[22TCSE206]

FOURTH SEMESTER B.Tech EXAMINATION, FEBRUARY-2024

ASSEMBLY C

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer all questions.

		PART-A	**05 X 02=10**		
1.	**a.**	List the basic data types in C language.	L-1	CO1	**02**
	b.	Write down the four basic components of an assembly language instruction.	L-2	CO1	**02**
	c.	Define the purpose of printf() and scanf() in C programming language.	L-2	CO1	**02**
	d.	List the number systems supported by computer architecture.	L-1	CO2	**02**
	e.	Write the four primary operations of MPU.	L-1	CO2	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Distinguish between Call-by-value and Call-by-reference with an example.	L-3	CO2	**05**
		OR			
	b.	Define Pointer. Discuss the process of declaring variable pointer variable.	L-2	CO1	**05**
3.	**a.**	Explain in brief the categories in which Computer Architecture is divided.	L-2	CO1	**05**
		OR			
	b.	Explain the difference between assembly language and high-level programming languages.	L-2	CO2	**05**
4.	**a.**	Write down the control structures in assembly language programming.	L-2	CO3	**05**
		OR			
	b.	Discuss the different types of Program Control Instructions.	L-2	CO3	**05**
5.	**a.**	Briefly discuss the shift and rotate instructions of 8086.	L-2	CO4	**05**
		OR			
	b.	Define the extended precision. Discuss the IEEE standard for numbers.	L-2	CO4	**05**
6.	**a.**	Define stored procedures. Write down different uses of it.	L-2	CO5	**05**
		OR			

	b.	Write an assembly language program to perform arithmetic addition operations using 8086 Microprocessor.	L-3	CO5	**05**
7.	**a.**	Discuss the benefits of multi-file programming.	L-2	CO6	**05**
		OR			
	b.	Discuss different methods of representing array with example.	L-2	CO6	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Define Structure in C. Explain structure definition with suitable example.	L-3	CO1	**10**
		OR			
	b.	Define Union. Write the declaration of it with example. Differentiate between Structure and Union.	L-3	CO1	**10**
9.	**a.**	Write a C program to find total number of alphabets, digits and special characters in a string.	L-3	CO2	**10**
		OR			
	b.	Explain the data transfer instructions.	L-2	CO2	**10**
10.	**a.**	Discuss different basic arithmetic operations in C with a suitable example.	L-3	CO3	**10**
		OR			
	b.	Explain the implementation of control flow in assembly language with a suitable example.	L-3	CO3	**10**
11.	**a.**	Describe the parsing of integer data in assembly C using suitable example.	L-3	CO4	**10**
		OR			
	b.	Discuss about different String Instructions used in assembly C.	L-2	CO5	**10**
12.	**a.**	Identify the crucial features to design the instruction set architecture for a specific purpose processor.	L-2	CO4	**10**
		OR			
	b.	Explain about compiling process and assembly process with a neat flow diagram?	L-2	CO5	**10**
13.	**a.**	Write about Interrupt and its types. Illustrate the phases of Interrupt Cycle with a neat flowchart.	L-2	CO6	**10**
		OR			
	b.	Discuss the memory hierarchy in computer system with regard to speed, size and cost. Explain about main memory and its types.	L-2	CO6	**10**